MA104 · DISCRETE MATHEMATICS

Assignment 02 — Membership Tables

Questions: **18**
Due: ___ / ___ / **2026**

Chapter 2 — Sets: set operations and set identities

Name: ..... **ID:** .....

Group: ..... **Date:** .....

Instructions. Work through the parts in order; each part builds on the one before it.

- Fill in **every** blank cell, including the intermediate columns — a final column on its own is not enough.
- Use **1** (the element is in the set) and **0** (it is not) only.
- Where a question asks whether two sets are equal, write **Yes** or **No** on the ruled line; for **No**, name a row in which the two sides differ.
- Hand-write your answers on this sheet. Write your name, ID and group above.
- **Answer without AI.** ChatGPT, Claude, Gemini and similar tools are not allowed, and neither is copying from a classmate. You may be asked to explain any answer at the board, and an answer you cannot explain will not be accepted.

Notation

Symbol	Name	Entry is 1 when ...
$A \cup B$	union	x is in A or in B (or both)
$A \cap B$	intersection	x is in A and in B
A^c	complement	x is not in A
$A - B$	difference	x is in A and not in B
$A \triangle B$	symmetric difference	x is in exactly one of A, B

Part A — Membership Tables for Concrete Sets

Lecture 2, Examples 1–4. One row per element of the universe U.

Question 1 cf. Ex. 1, 2, 4, 6

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 5, 8\}$ and $B = \{2, 4, 5, 6\}$. Complete the membership table, then write each set in roster form.

Element	A	B	$A \cup B$	$A \cap B$	$A - B$	$B - A$	$A \triangle B$
1							
2							
3							
4							
5							
6							
7							
8							

$A \cup B$ = ... $A \cap B$ = ...

$A - B$ = ... $B - A$ = ...

$A \triangle B$ = ...

Question 2 cf. Ex. 3

Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 3, 4, 7\}$, $B = \{2, 3, 5, 7\}$ and $C = \{3, 4, 5, 6\}$. Complete the membership table, then write each set in roster form.

Element	A	B	C	$B \cap C$	$A \cup (B \cap C)$	A^c	$A^c \cap C$
1							
2							
3							
4							
5							
6							
7							

$A \cup (B \cap C)$ = ... $A^c \cap C$ = ...

Question 3 cf. Ex. 3, 7

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 5\}$ and $B = \{2, 3\}$. Complete the membership table, write the two sets in roster form, and say what you notice.

Element	A	B	$A \cup B$	$(A \cup B)^c$	A^c	B^c	$A^c \cap B^c$
1							
2							
3							
4							
5							
6							

$(A \cup B)^c$ = ... $A^c \cap B^c$ = ...

What do you notice? .....

Part B — Proving or Disproving Set Identities

Lecture 2, Examples 5–15. Build a column for each side; the identity holds exactly when the two final columns agree in every row.

Question 4 cf. Ex. 9

Use a membership table to decide whether the identity holds:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	_____	_____	_____	_____	_____
1	1	0	_____	_____	_____	_____	_____
1	0	1	_____	_____	_____	_____	_____
1	0	0	_____	_____	_____	_____	_____
0	1	1	_____	_____	_____	_____	_____
0	1	0	_____	_____	_____	_____	_____
0	0	1	_____	_____	_____	_____	_____
0	0	0	_____	_____	_____	_____	_____

Equal? .....

Question 5 cf. Ex. 12, 13

Use a membership table to decide whether the identity holds:

$$A \cap (B - C) = (A \cap B) - (A \cap C)$$

A	B	C	$B - C$	$A \cap (B - C)$	$A \cap B$	$A \cap C$	$(A \cap B) - (A \cap C)$
1	1	1	_____	_____	_____	_____	_____
1	1	0	_____	_____	_____	_____	_____
1	0	1	_____	_____	_____	_____	_____
1	0	0	_____	_____	_____	_____	_____
0	1	1	_____	_____	_____	_____	_____
0	1	0	_____	_____	_____	_____	_____
0	0	1	_____	_____	_____	_____	_____
0	0	0	_____	_____	_____	_____	_____

Equal? .....

Question 6 cf. Ex. 5, 10

Use a membership table to decide whether the identity holds:

$$(A - B) \cup B = A$$

A	B	$A - B$	$(A - B) \cup B$	A
1	1	_____	_____	_____
1	0	_____	_____	_____
0	1	_____	_____	_____
0	0	_____	_____	_____

Equal? .....

Question 7 cf. Ex. 14, 15

Use a membership table to decide whether the identity holds:

$$A - (B - C) = (A - B) - C$$

A	B	C	$B - C$	$A - (B - C)$	$A - B$	$(A - B) - C$
1	1	1				
1	1	0				
1	0	1				
1	0	0				
0	1	1				
0	1	0				
0	0	1				
0	0	0				

Equal? .....

Question 8 cf. Ex. 6, 14

Use a membership table to decide whether the identity holds:

$$A \triangle (B \triangle C) = (A \triangle B) \triangle C$$

A	B	C	$B \triangle C$	$A \triangle (B \triangle C)$	$A \triangle B$	$(A \triangle B) \triangle C$
1	1	1				
1	1	0				
1	0	1				
1	0	0				
0	1	1				
0	1	0				
0	0	1				
0	0	0				

Equal? .....

Part C — Unsolved Exercises on Membership Tables

The Assignment 2 exercise sheet, Exercises 1–10.

Question 9 Ex. 1

Complete the membership table for:

$$A \cup B^c$$

A	B	B^c	$A \cup B^c$
1	1	_____	_____
1	0	_____	_____
0	1	_____	_____
0	0	_____	_____

Question 10 Ex. 2

Complete the membership table for:

$$A^c \cap B$$

A	B	A^c	$A^c \cap B$
1	1	_____	_____
1	0	_____	_____
0	1	_____	_____
0	0	_____	_____

Question 11 Ex. 3

Complete the membership table for:

$$(A - B)^c$$

A	B	$A - B$	$(A - B)^c$
1	1	_____	_____
1	0	_____	_____
0	1	_____	_____
0	0	_____	_____

Question 12 Ex. 4

Complete the membership table for:

$$(A \cup B) \cap A^c$$

A	B	$A \cup B$	A^c	$(A \cup B) \cap A^c$
1	1	_____	_____	_____
1	0	_____	_____	_____
0	1	_____	_____	_____
0	0	_____	_____	_____

Question 13 Ex. 5

Complete the membership table for:

$$(A \cap B) \cup A^c$$

A	B	$A \cap B$	A^c	$(A \cap B) \cup A^c$
1	1	_____	_____	_____
1	0	_____	_____	_____
0	1	_____	_____	_____
0	0	_____	_____	_____

Question 14 Ex. 6

Complete the membership table for:

$$(A \cup B) - (A \cap B)$$

A	B	$A \cup B$	$A \cap B$	$(A \cup B) - (A \cap B)$
1	1	_____	_____	_____
1	0	_____	_____	_____
0	1	_____	_____	_____
0	0	_____	_____	_____

Question 15 Ex. 7

Complete the membership table for:

$$(A \cap B^c) \cup (A^c \cap B)$$

A	B	$A \cap B^c$	$A^c \cap B$	$(A \cap B^c) \cup (A^c \cap B)$
1	1	_____	_____	_____
1	0	_____	_____	_____
0	1	_____	_____	_____
0	0	_____	_____	_____

Question 16 Ex. 8

Complete the membership table for:

$$(A \cup B) \cap C$$

A	B	C	$A \cup B$	$(A \cup B) \cap C$
1	1	1	_____	_____
1	1	0	_____	_____
1	0	1	_____	_____
1	0	0	_____	_____
0	1	1	_____	_____
0	1	0	_____	_____
0	0	1	_____	_____
0	0	0	_____	_____

Question 17 Ex. 9

Complete the membership table for:

$$A \cup (B \cap C)$$

A	B	C	$B \cap C$	$A \cup (B \cap C)$
1	1	1	_____	_____
1	1	0	_____	_____
1	0	1	_____	_____
1	0	0	_____	_____
0	1	1	_____	_____
0	1	0	_____	_____
0	0	1	_____	_____
0	0	0	_____	_____

Question 18 Ex. 10

Complete the membership table for:

$$(A - B) \cup C$$

A	B	C	$A - B$	$(A - B) \cup C$
1	1	1	_____	_____
1	1	0	_____	_____
1	0	1	_____	_____
1	0	0	_____	_____
0	1	1	_____	_____
0	1	0	_____	_____
0	0	1	_____	_____
0	0	0	_____	_____

— END OF ASSIGNMENT —